



Territory Briefing: Catania

Professional province-level screening based on ARCUS historical evidence, territorial exposure, ARCUS/AINOP Collapse Rate and historical-pattern reading.

PRIORITY

29/100

COLLAPSE

0.54x / national rank 51

EVENTS

1

DECISION MESSAGE

Concentrated critical event

REPORT DATE

Jun 10, 2026

CONTEXT

Bridge

PROVINCE

Catania

OUTPUT

New construction

ARCUS is built on the first systematic database of collapsed bridges in Italy from 2000 to today: the report does not merely rank cases, it reads documented historical patterns to guide preliminary checks.

The selected province requires preliminary screening for Bridge interventions. The value is not the case list: it is understanding whether Catania has already shown concentrated vulnerability under extreme conditions or risk distributed over time.

RECOMMENDED USE

ARCUS reading: for new Bridge interventions, the Material signal does not define a design solution. It identifies the technical attention fields that should be checked before design, due diligence or site-specific investigation decisions in the province of Catania.

PRIORITY INDEX

29/100

exposure index + heritage reading

COLLAPSE RATE

0.54x

ARCUS/AINOP national benchmark; confidence high

HISTORICAL EVENTS

1

Material dominant

RELIABILITY

81/100

2 linked sources

COLLAPSE RATE INTERPRETATION

The 0.54x rate versus the national average places Catania among the provinces with documented historical vulnerability above the ARCUS/AINOP benchmark. Catania is national rank 51 by documented ARCUS/AINOP collapse rate. Collapse Rate confidence: high. The AINOP denominator should be read as available bridge-census coverage, not as a perfect inventory of the bridge stock.



CATANIA / ARCUS ATLAS EXTRACT

GEOMORPHOLOGY AND CONTEXT

Catania shows a heterogeneous territorial profile: 1 ARCUS events across 1 municipalities indicate that the structural driver must be read together with local morphology, drainage network, slopes and crossing continuity.

MAP READING

The map locates 1 ARCUS cases across 1 municipalities and should be read with the historical pattern: Concentrated critical event. Dominant date: 2011-05-08.

HISTORICAL PATTERN

Concentrated critical event. collapses concentrated in a single critical-event scenario (2011-05-08, 100%).

DOMINANT TRIGGER**100%**

Material

HAZARD**structural**

dominant territorial signal

READING**single event**

100% in one flood scenario; check extreme single-event design scenarios.

- Use the territorial exposure indicators to identify the dominant territorial signal.
- Review the documented concentrated historical scenario and use the map/appendix only as case reference.
- Request the minimum technical data package for the dominant structural signal: local hazard study, asset geometry, foundation/abutment exposure, inspection history and source-backed site constraints.
- Commission site-specific checks calibrated to the dominant structural context and local geomorphology before moving from provincial screening to design assumptions.
- Use CSV and GeoJSON exports for technical coordination.

DATA REQUEST PACKAGE

Local hazard study; asset geometry and foundations; inspection history; source-backed site constraints.

COLLAPSE RATE CONFIDENCE NOTE

Collapse Rate confidence: high. The AINOP denominator should be read as available bridge-census coverage, not as a perfect inventory of the bridge stock.

The following list is a documentary reference, not an operational P1/P2/P3 ranking. The decision message remains the provincial historical pattern.

ID	Date	Municipality	Severity	Cause
B11.05.01	May 08, 2011	Caltagirone	Total Collapse	Material

Layer	Meaning
Priority Index	Territorial exposure index: synthesizes dominant hazard, ARCUS case density, event concentration and evidence reliability. It does not directly measure historical bridge-stock vulnerability.
Collapse Rate	Ratio between ARCUS cases and AINOP counted bridges in the province. Value: 0.54x, national rank 51; it should be read separately from the Priority Index.
Why Separate	Priority Index indicates where the territory has elevated hazard exposure; Collapse Rate indicates where the bridge stock has shown historical vulnerability above or below the national average.
Historical Failure Context	Historical-pattern reading: concentrated event or distributed risk over time.
Hydraulic Cause Category	Hydraulic cause category: ARCUS uses 'hydraulic' as the documented trigger when sources confirm a flood or high-water event as the proximate cause. Mechanism-level classification - scour, bank erosion, debris impact, overtopping - is applied only where primary technical sources provide
Evidence Window	Observed period: ARCUS database of collapsed bridges in Italy from 2000 to today; close events in the same time window are read as one scenario, not decision duplicates.
AINOP Coverage	Collapse Rate uses AINOP counted bridges as provincial denominator. Confidence: high; local undercounting may inflate the ratio.
Evidence Classes	A = primary institutional/technical source; B = reliable technical or media source; C = generic documentary evidence; D = weak evidence to strengthen.